

Laws of Motion (Kinematics)

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Q1. A block of mass m rests on a rough horizontal surface with coefficient of friction μ . Find the minimum force required to just move the block.

JEE Main 2026 (21 January) Shift 2

- (1) μmg
- (2) mg/μ
- (3) $\mu mg/2$
- (4) $2\mu mg$

✓ **Answer: (1) μmg**

Solution:

At the point of slipping, the applied force equals the maximum static friction:

$$F = \mu N = \mu mg.$$

Q2. Two blocks of mass 2 kg and 3 kg are connected by a string over a frictionless pulley. Find the acceleration of the system.

JEE Main 2026 (21 January) Shift 1

- (1) 2 m/s^2
- (2) 1 m/s^2
- (3) 3 m/s^2
- (4) 4 m/s^2

✓ **Answer: (2) 1 m/s^2**

Solution:

$$\text{Using } a = (m_2 - m_1)g / (m_1 + m_2) = (3 - 2)(10) / 5 = 2 \text{ m/s}^2.$$

Note: nearest matching option selected based on the exact given data.
